

3月25日 周三

微积分，共5题。

木头被做成了纸，纸片折叠成为树。纸折成的大家，虚假之中仍有一丝真我。

祝顺利！

$$1. \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{x(1 - \cos \sqrt{x})}$$

$$A = \lim_{x \rightarrow 0} \frac{1 - \frac{1}{2}(\cos x - 1) - 1}{\frac{1}{2}x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{2}(1 - \frac{x^2}{2})}{\frac{1}{2}x^2} = \frac{1}{2}$$

$$2. \int \sqrt{x^2 \pm a^2} dx$$

$$(1) I_1 = \int \sqrt{x^2 + a^2} dx \quad \begin{matrix} x = a \tan t \\ dx = a \sec^2 t dt \end{matrix} \quad \int a^2 \sec^3 t dt = a^2 \int \sec t \cdot d \tan t = a^2 \sec t \tan t - a^2 \int \sec t (\sec^2 t - 1) dt.$$

$$\text{故 } I = \frac{1}{2} a^2 \sec t \tan t + \frac{1}{2} a^2 \int \sec t dt, \quad \int \sec t dt = \int \frac{1}{\cos t} dt = \int \frac{ds}{1-s^2} = \frac{1}{2} \int \left(\frac{1}{1-s} + \frac{1}{1+s} \right) ds.$$

$$\text{故 } I = \frac{1}{2} a^2 \sec t \tan t + \frac{a^2}{2} \ln \left| \sec t + \tan t \right| + C = \frac{1}{2} a^2 \ln \left| \frac{1+s}{1-s} \right| = \frac{1}{2} a^2 \ln \left| \sec t + \tan t \right| + C$$

$$= \frac{a^2}{2} \frac{\sqrt{x^2 + a^2}}{a} \cdot \frac{x}{a} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$= \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C.$$

$$(2) \text{ 对于 } I_2 = \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + C.$$

$$3. \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{x(1 - \cos \sqrt{x})}$$

$$\text{由 1. 得, } A = \frac{1}{2}$$

4. 设 $y = f(x)$ 由 $\begin{cases} x = t^2 + 2t \\ t^2 - y + a \sin y = 1 \end{cases}$ 确定, 若 $y(0) = b$, 求 $\frac{d^2y}{dx^2} \Big|_{t=0}$

$$y(0)=b, \quad a \sin b = 1+b$$

$$\text{有 } 0 = 2t - y' + ay' \cos y, \quad y' = \frac{2t}{1 - a \cos y}, \quad x' = 2t + 2$$

$$\frac{dy}{dx} = \frac{t}{(1 - a \cos y)(t+1)} \Big|_{t=0} = 0$$

$$\frac{d^2y}{dx^2} = \frac{(dy/dx)'_t}{x'} = \frac{[(1 - a \cos y)(t+1) - t(a y' \sin y (t+1) + 1 - a \cos y)]}{(1 - a \cos y)^2 (t+1)^2 (2t+2)}$$

$$\frac{d^2y}{dx^2} \Big|_{t=0} = (1 - a \cos b) / (1 - a \cos b)^2 \cdot 2 = \frac{1}{2(1 - a \cos b)}$$

5. 设 $f(x) = x, x \geq 0, g(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & x > \frac{\pi}{2} \end{cases}$, 求 $\Phi(x) = \int_0^x f(t)g(x-t)dt$ 的表达式 ($x \geq 0$)

$$\Phi(x) = \int_0^x f(t)g(x-t)dt \xrightarrow[-dt=du]{x-t=u} \int_0^x f(x-u)g(u)du = \int_0^x (x-u)g(u)du.$$

$$(i) \quad 0 \leq x \leq \frac{\pi}{2} \quad \text{有} \quad \int_0^x (x-u)g(u)du = \int_0^x (x-u)\sin u du = -(x-u)\cos u - \sin u \Big|_0^x \\ = -\sin x + x$$

$$(ii) \quad x > \frac{\pi}{2} \quad \text{有} \quad \int_0^x (x-u)g(u)du = \int_0^{\frac{\pi}{2}} (x-u)\sin u du = -(x-u)\cos u \Big|_0^{\frac{\pi}{2}} - 1 \\ = x - 1$$

$$\text{综上 } \Phi(x) = \begin{cases} x - \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ x - 1, & x > \frac{\pi}{2} \end{cases}$$

